

Aug 25, 2022

Introduction to Computer and Network Security

Part 1

Question 1 [40 points]

In the context of access control, describe the general architecture of an access control enforcement mechanism including (a) subjects, (b) requests, (c) guard, (d) policy, (e) isolation boundaries, (f) audit log, and (g) the role of authentication and authorization.

Question 2 [10 points]

Define the notion of cryptosystem.

Part 2

Question 3 [35 points]

In the context of data protection and privacy, (a) give two complementary definitions of privacy and briefly comment on them, (b) explain the linkage attack on anonymized dataset with an example, and (c) explain the trade-off between anonymity and utility in the context of realizing data sets.

Question 4 [15 points]

Explain the framework of Attribute Based Access Control (ABAC), in particular how an access request is processed, and the reasons for which it has been introduced with respect to previous models (such as RBAC).

Please, remember...

On each sheet, make sure to write:

Name

Surname

Student ID number

Question number